

X-Ray Image Classification System

AI Project Documentation

X-Ray Image Classification System

Module of Smart Hospital Management System

1. Project Title

AI-Based X-Ray Image Classification System Using Deep Learning

2. Project Overview

The **X-Ray Image Classification System** is an Artificial Intelligence (AI)-based healthcare module designed to analyze medical X-ray images and classify them automatically using Deep Learning techniques.

The system uses Computer Vision and Deep Learning models to identify abnormalities in musculoskeletal X-ray images. The project uses the **MURA v1.1 (Musculoskeletal Radiographs) dataset** and applies advanced AI architectures including:

- MobileNetV3 (CNN-based approach)
- Vision Transformer (Transformer-based approach)
- Swin Transformer (optional advanced approach)

The purpose of this system is to assist doctors and radiologists by providing faster preliminary analysis of X-ray images. The AI model generates classification results with confidence scores, allowing healthcare professionals to review predictions before making final medical decisions.

The system will be integrated as an AI module inside the **Smart Hospital Management System**.

3. Problem Statement

Medical image analysis requires significant expertise and time from radiologists. Hospitals handle a large number of X-ray images daily, which increases workload and delays diagnosis.

Traditional manual analysis can be affected by:

- High workload
- Human fatigue
- Time constraints
- Variations between specialists

This project aims to develop an AI-assisted X-ray classification system that can automatically analyze X-ray images and identify possible abnormalities.

The system will support healthcare professionals by providing faster screening and decision assistance.

4. Project Goals

The main goal of this project is to develop an AI-powered X-ray classification system capable of:

- Processing medical X-ray images.
 - Extracting important visual features.
 - Classifying X-ray images automatically.
 - Detecting abnormal cases.
 - Comparing CNN and Transformer-based approaches.
 - Providing prediction confidence scores.
 - Supporting doctors with AI-assisted analysis.
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5. Dataset Information

Dataset Name

MURA v1.1

(Musculoskeletal Radiographs Dataset)

Dataset Source

Available on Kaggle.

Dataset Description

MURA is one of the largest public datasets for musculoskeletal radiograph analysis.

It contains:

- More than 40,000 X-ray images.
 - More than 14,000 studies.
 - Multiple body regions.
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Supported Body Parts

The dataset contains X-rays of:

- Elbow
 - Finger
 - Forearm
 - Hand
 - Humerus
 - Shoulder
 - Wrist
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Dataset Classes

The system performs binary classification:

Normal

X-ray image does not contain abnormal findings.

Abnormal

X-ray image contains possible abnormalities.

Dataset Format

Input:

Image Files

.jpg

.jpeg

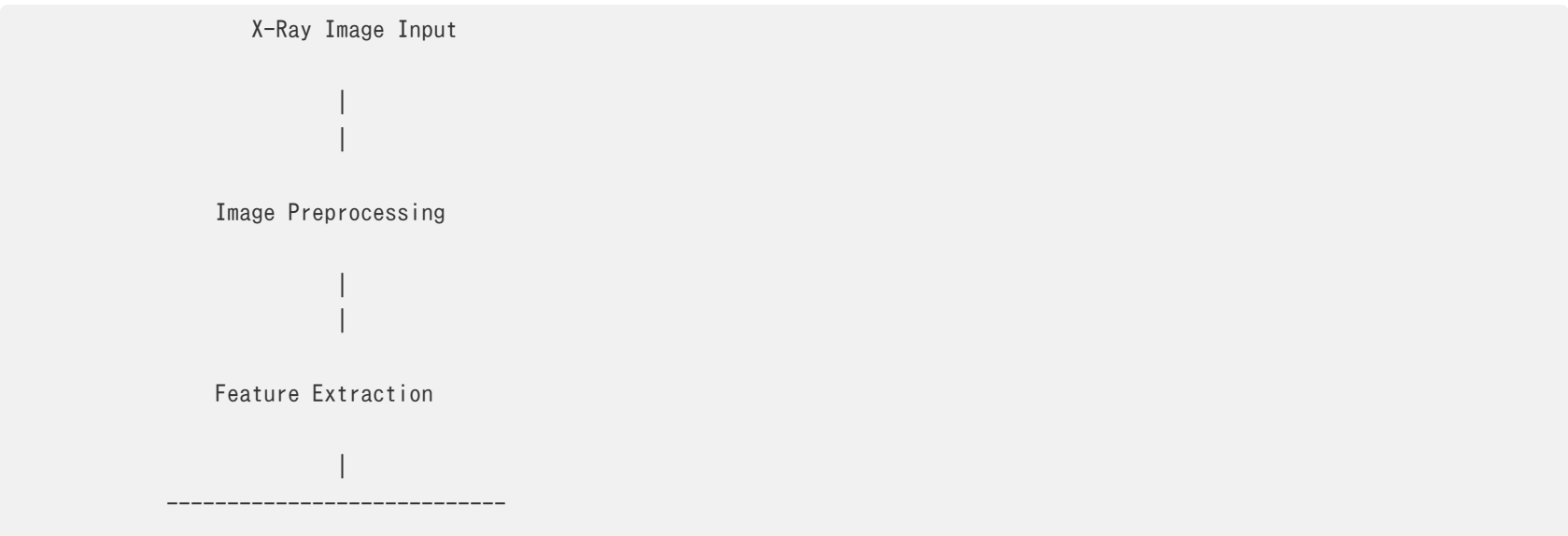
.png

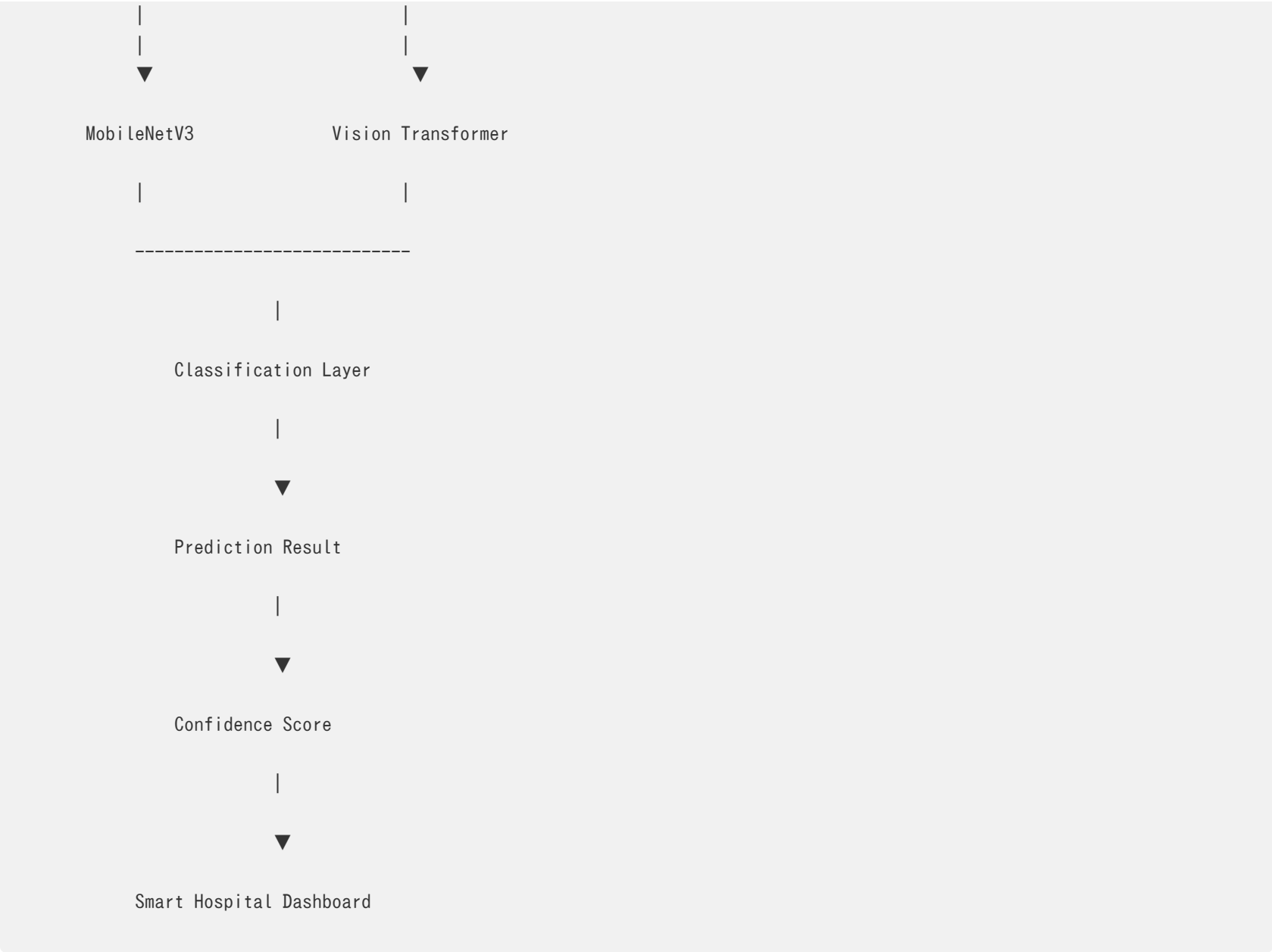
6. Project Objectives

The objectives of this project are:

1. Develop an AI-based X-ray classification system.
2. Apply Computer Vision techniques for medical image analysis.
3. Implement CNN-based classification using MobileNet.
4. Implement Transformer-based classification using Vision Transformer.
5. Compare CNN and Transformer performance.
6. Evaluate models using standard evaluation metrics.
7. Generate confidence-based predictions.
8. Integrate the AI model into Smart Hospital Management System.

7. System Architecture





8. System Workflow

Step 1: X-Ray Image Input

The system accepts medical X-ray images uploaded by users.

Supported formats:

- PNG
- JPG
- JPEG

Example:

Input:

Patient X-Ray Image

Step 2: Image Preprocessing

Medical images require preprocessing before being passed to AI models.

Preprocessing operations include:

Image Resizing

Original images may have different dimensions.

Example:

Before:

1024 × 1024 pixels

After:

224 × 224 pixels

Normalization

Pixel values are converted into a suitable range.

Example:

Before:

0 - 255

After:

0 - 1

Image Enhancement

Techniques:

- Contrast adjustment
- Noise reduction
- Image sharpening

Step 3: Data Augmentation

To improve model performance, training images are augmented.

Techniques:

- Rotation
- Zoom
- Horizontal flipping
- Brightness adjustment
- Contrast modification
- Random cropping

Benefits:

Prevents overfitting

- Improves generalization
- Creates additional training variations

9. Deep Learning Models

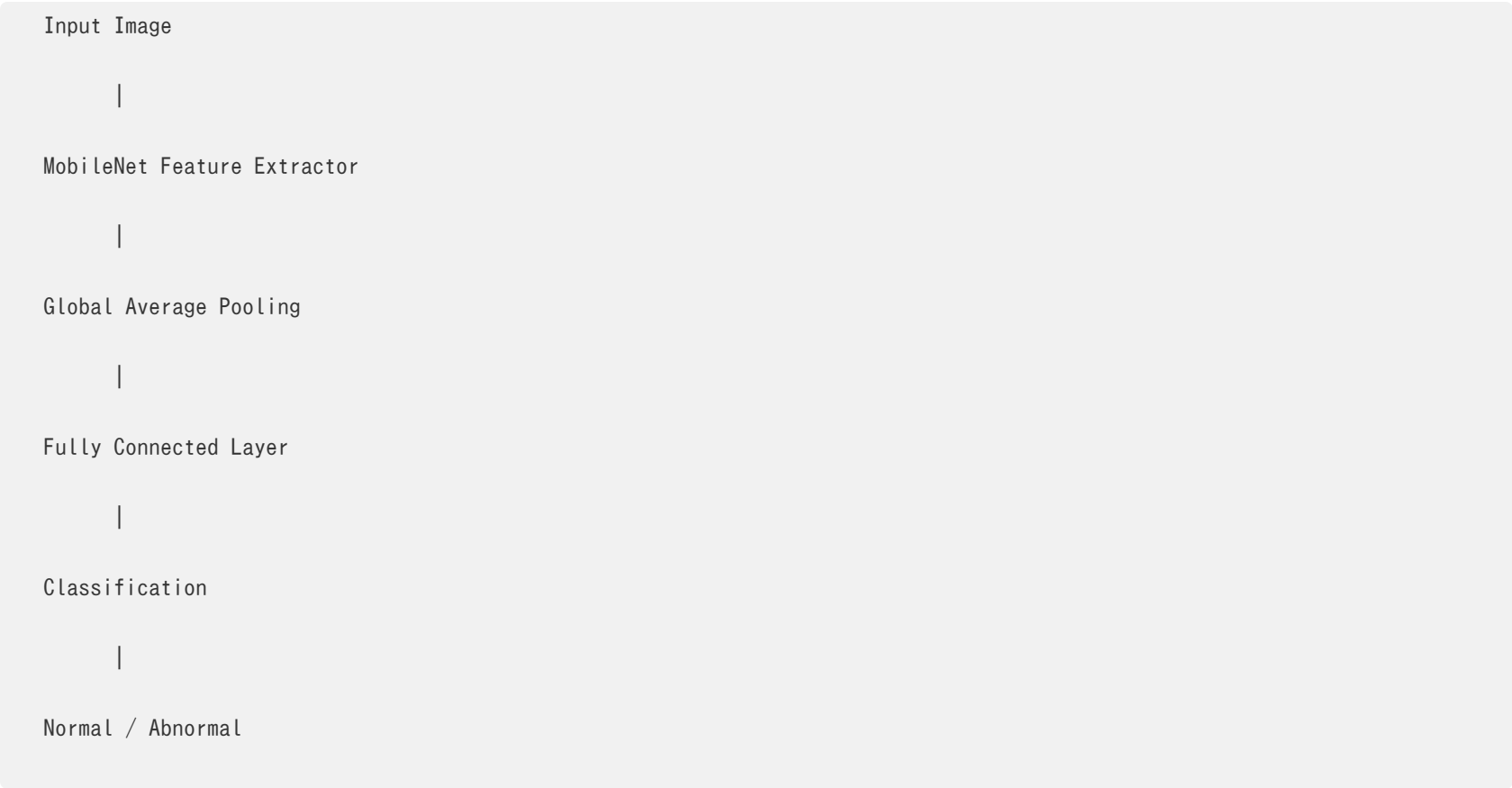
The project implements two different AI approaches.

Model 1: MobileNetV3

Overview

MobileNetV3 is a lightweight Convolutional Neural Network designed for efficient image classification. It is suitable for healthcare applications where fast inference is required.

Architecture



Advantages

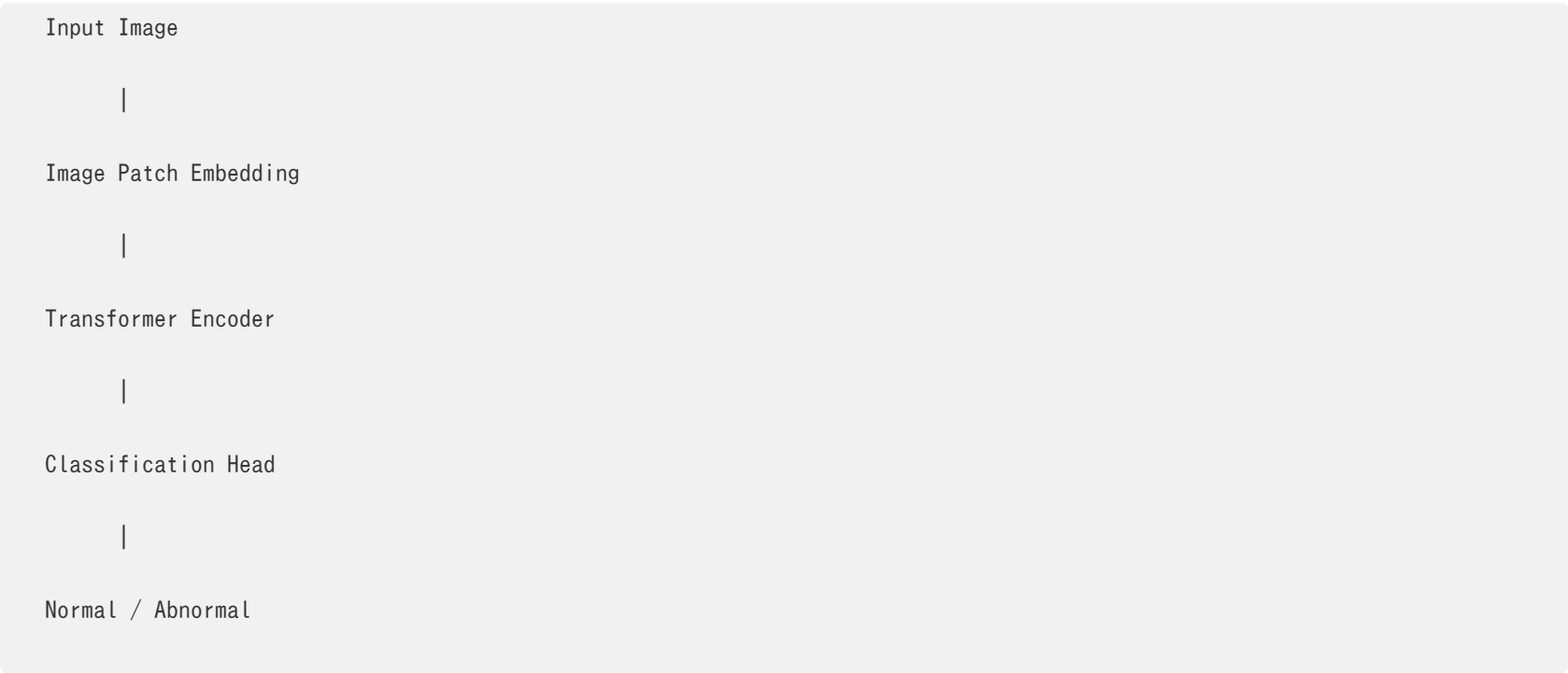
- Lightweight architecture.
- Fast training.
- Low computational requirements.
- Suitable for real-time applications.
- Works efficiently on limited hardware.

Model 2: Vision Transformer (ViT)

Overview

Vision Transformer applies Transformer architecture to image classification. Instead of convolution operations, it divides images into patches and learns relationships between different image regions.

Architecture



Advantages

- Learns global image patterns.
- Captures complex relationships.
- Strong performance on large datasets.
- Modern state-of-the-art architecture.

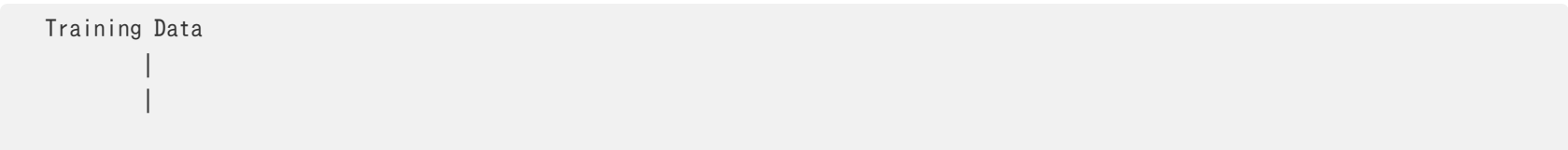
10. Model Training Process

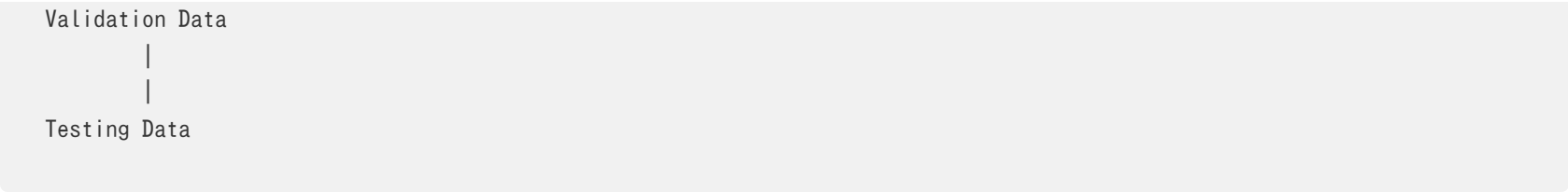
Dataset Preparation

Steps:

1. Collect MURA dataset.
2. Organize images.
3. Apply preprocessing.
4. Create training and testing sets.

Dataset Split:





Training Configuration

Example settings:

Parameter	Value
Image Size	224 × 224
Batch Size	32
Epochs	30-50
Optimizer	Adam
Learning Rate	0.0001
Loss Function	Binary Cross Entropy

11. Model Evaluation

The trained models are evaluated using:

Accuracy

Measures overall correct predictions.

Formula:

Correct Predictions / Total Predictions

Precision

Measures correctness of abnormal predictions.

Recall

Measures how many abnormal cases were detected.

F1 Score

Combination of precision and recall.

Confusion Matrix

Shows:

- True Positive
- True Negative
- False Positive
- False Negative

ROC-AUC Score

Measures classification ability across different thresholds.

12. Prediction System

Input

Patient X-ray image.

Example:

Shoulder X-Ray

AI Processing

The model analyzes:

- Image features
- Patterns
- Abnormal characteristics

Output

Example:

Prediction:

Abnormal

Confidence:

96.5%

Probability:

Abnormal :
96.5%

Normal :
3.5%

13. Technology Stack

Programming Language

Python

Deep Learning Frameworks

- TensorFlow
 - PyTorch
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Computer Vision

- OpenCV
 - Pillow
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Data Processing

- NumPy
 - Pandas
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Visualization

- Matplotlib
 - Seaborn
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Machine Learning

- Scikit-Learn
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AI Models

- MobileNetV3
 - Vision Transformer
 - Swin Transformer
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Backend Deployment

- Flask
 - FastAPI
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Database Integration (Optional)

- PostgreSQL

- MySQL
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14. Project Development Phases

Phase 1: Research and Planning

Tasks:

- Understand medical imaging.
- Study X-ray classification.
- Analyze MURA dataset.
- Study AI architectures.

Deliverables:

- Research documentation.
 - Technical understanding.
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Phase 2: Dataset Preparation

Tasks:

- Download dataset.
- Organize images.
- Perform preprocessing.
- Apply augmentation.

Deliverables:

- Prepared dataset.
 - Data pipeline.
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Phase 3: Model Development

Tasks:

- Implement MobileNet model.
- Implement Transformer model.
- Train models.
- Tune parameters.

Deliverables:

- Trained AI models.
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Phase 4: Evaluation

Tasks:

- Test models.
- Generate metrics.
- Compare performance.

Deliverables:

- Evaluation report.
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Phase 5: Deployment

Tasks:

- Create prediction API.
- Build image upload system.
- Integrate with hospital system.

Deliverables:

- Working AI module.
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15. Project Responsibilities

All team members will participate in all project phases.

Responsibilities include:

Research

- Study AI healthcare applications.
 - Analyze existing solutions.
 - Review research papers.
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Dataset Management

- Prepare dataset.
 - Perform preprocessing.
 - Manage training data.
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Model Development

- Implement AI architectures.
 - Train deep learning models.
 - Improve performance.
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Testing

- Test predictions.

- Analyze results.
 - Validate accuracy.
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Deployment

- Develop API.
 - Integrate AI model.
 - Create user interface.
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Documentation

- Prepare reports.
 - Record experiments.
 - Create presentation.
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16. System Limitations

The system has some limitations:

- AI predictions are not final medical diagnoses.
 - Performance depends on dataset quality.
 - Poor image quality can reduce accuracy.
 - Requires validation before clinical usage.
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17. Future Improvements

Future enhancements include:

- Multi-disease classification.
 - Bone fracture detection.
 - Explainable AI using Grad-CAM.
 - DICOM image support.
 - Medical report generation using LLM.
 - PACS integration.
 - Real-time hospital deployment.
 - Federated learning for patient privacy.
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18. Expected Final Deliverables

The final project will include:

- ✓ AI X-ray classification model
- ✓ MobileNet implementation

- ✓ Vision Transformer implementation
 - ✓ Dataset preprocessing pipeline
 - ✓ Model evaluation report
 - ✓ Prediction API
 - ✓ Smart Hospital integration
 - ✓ Technical documentation
 - ✓ Final presentation
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19. Final Project Outcome

At the completion of this project, the team will have developed an AI-powered X-ray classification module capable of:

- Automatically analyzing medical X-ray images.
 - Detecting abnormal cases.
 - Comparing CNN and Transformer approaches.
 - Generating confidence-based predictions.
 - Supporting doctors in faster medical image screening.
 - Integrating Artificial Intelligence into Smart Hospital Management Systems.
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Project Status: AI Research + Development Module

Domain: Artificial Intelligence / Healthcare / Computer Vision

Dataset: MURA v1.1

Primary Models: MobileNetV3 + Vision Transformer